2D Plane Stress Analysis and Evaluation of Stress Concentration in a Plate with a Circular Hole

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1 Introduction

Stress concentration is a critical phenomenon in solid mechanics where local stresses increase significantly due to abrupt changes in geometry, such as a central circular hole in a structural plate. Accurately predicting these peak stresses is essential to prevent premature fatigue failure in mechanical components.

Following the establishment of the mathematical framework in Phase 1, the objective of Phase 2 is to translate the analytical 2D plane-stress formulations into a procedural MATLAB computational solver. Furthermore, this report details a rigorous Grid Convergence Study to mathematically validate the accuracy and stability of the custom solver across varying mesh densities.

1.1 Problem Geometry

To optimize computational efficiency, a quarter-symmetry model is utilized. The physical domain represents the top-right quarter of the plate, with dimensions of length $L = 50$ mm, width $W = 25$ mm, and a hole radius of $a = 20$ mm. A uniform uniaxial tensile stress of $\sigma_0 = 100$ MPa is applied to the far right edge.

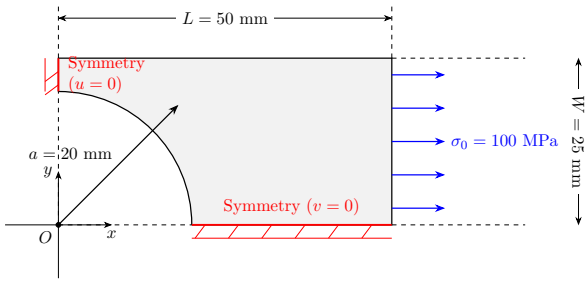


Figure 1: Physical domain of the quarter-symmetry computational model. To optimize solver efficiency and reduce computational overhead, the simulation is not performed on the full-scale plate. Instead, only the top-right quarter is modeled, with exact kinematic symmetry boundaries enforced along the $x = 0$ and $y = 0$ axes.

2 Theoretical Formulation

The foundation of the finite element solver relies on deriving the element stiffness matrix, which connects nodal displacements to applied forces via the Virtual Work principle.

2.1 Constitutive Material Matrix

Assuming the plate is thin relative to its in-plane dimensions, we apply the 2D plane stress assumption ($\sigma_{zz} = \tau_{xz} = \tau_{yz} = 0$). The constitutive material matrix, D, relates the in-plane stress vector σ to the strain vector ε via the generalized Hooke's law:

$$D = \frac{E}{1-\nu^2}\begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \tag{1}$$

where $E = 210,000$ MPa (Steel) and $\nu = 0.3$.

2.2 Q4 Element Formulation

We utilize the 4-node isoparametric quadrilateral (Q4) element. The physical coordinates (x, y) are mapped to natural coordinates (ξ, η). The bilinear shape functions are:

$$N_i(\xi, \eta) = \frac{1}{4}(1 + \xi_i\xi)(1 + \eta_i\eta) \tag{2}$$

The strain vector is derived from spatial derivatives: $\varepsilon = Bd_e$. To obtain physical derivatives from the natural domain, we apply the Jacobian matrix, J:

$$J = \begin{bmatrix} \sum \frac{\partial N_i}{\partial \xi} x_i & \sum \frac{\partial N_i}{\partial \xi} y_i \\ \sum \frac{\partial N_i}{\partial \eta} x_i & \sum \frac{\partial N_i}{\partial \eta} y_i \end{bmatrix} \tag{3}$$

Applying the principle of virtual work, the computable elemental stiffness matrix is the area integral evaluated via Gauss-Legendre quadrature:

$$K_e = t \int_{-1}^{1} \int_{-1}^{1} B^T D B \det(J) \, d\xi \, d\eta \tag{4}$$

3 Meshing and Discretization

The domain discretization was performed using Gmsh, an open-source 2D/3D finite element mesh generator. A custom geometry script (`.geo`) was written to define the points, lines, and the quarter-circle arc.

To accurately capture the steep stress gradients near the circular void while minimizing computational cost elsewhere, local mesh refinement was enforced at the hole boundary by dividing the characteristic length (lc) by a factor of 3. The `Recombine Surface` algorithm was utilized to force the generation of strictly Q4 quadrilateral elements over the default triangular elements. The resulting structured mesh data was exported in the Version 2 ASCII `.msh` format.

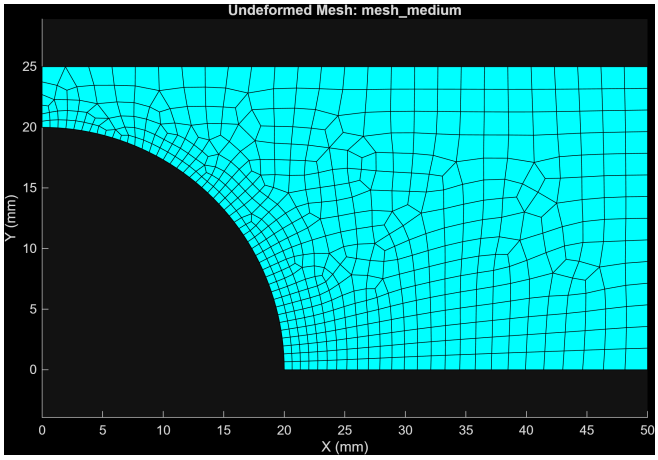


Figure 2: Representative Q4 mesh generated via Gmsh, illustrating heavy refinement near the hole boundary.

4 MATLAB Implementation Details

The theoretical mathematics were translated into a modular MATLAB script. This section outlines the logic connecting the continuous solid mechanics equations to discrete programmatic arrays, ensuring robust memory management and efficient evaluation.

4.1 Memory Allocation and Initialization

After a custom parser reads the Gmsh text file to construct the `NodeCoords` and `Connectivity` arrays, the global matrices are initialized. Because the global stiffness matrix K is exceptionally large but sparse, it is critical to allocate it via MATLAB's `sparse()` function to prevent RAM overflow. The 2×2 Gauss-Legendre quadrature points are also initialized here.

```
totalDOFs = 2 * numNodes;
K = sparse(totalDOFs, totalDOFs);
F = zeros(totalDOFs, 1);

gaussPts = [-1/sqrt(3), 1/sqrt(3)];
weights = [1, 1];
```

4.2 The Element Assembly Loop

The core engine iteratively evaluates every Q4 element. The numerical integration routine computes the Jacobian matrix and its inverse to form the strain-displacement matrix (B). The 8×8 local stiffness matrix (K_e) is accumulated and mapped into the global K matrix. Crucially, a pristine copy of K is saved prior to enforcing boundary conditions to ensure accurate thermodynamic calculations later.

```
for e = 1:numElems
    % ... [Coordinate extraction & B-matrix assembly] ...

    % Add contribution to local stiffness matrix
    Ke = Ke + (B' * D * B) * detJ * w_i * w_j * t;

    % Map to global sparse matrix
    K(sctr, sctr) = K(sctr, sctr) + Ke;
end

% CRITICAL: Save unpenalized K for true energy calculation
K_orig = K;
```

4.3 Linear Solver and Global Metrics

The quarter-symmetry essential boundary conditions are enforced using the Penalty Method. A massive scalar (10^{15}) is added to the main diagonal at constrained nodes. The system is timed and solved via the backslash operator. Following the solution, global kinematic and thermodynamic metrics—such as Maximum X-Displacement and Total Strain Energy—are evaluated.

```
% Penalty Method
penalty_value = 1e15;
K(fixed_DOFs, fixed_DOFs) = K(fixed_DOFs, fixed_DOFs) + penalty_value * speye(length(fixed_DOFs));

tic; % Start solver timer
U = K \ F;
solve_time = toc; % Stop solver timer

% Calculate Global Convergence Metrics
U_x = U(1:2:end);
max_u = max(U_x); % Max X-displacement

% Strain Energy = 1/2 * U^T * K_original * U
strain_energy = 0.5 * dot(U, K_orig * U);
```

4.4 Stress Recovery

To recover the continuous stress field from the nodal displacements, the strain ($\varepsilon = Bd_e$) and stress ($\sigma = D\varepsilon$) vectors are evaluated directly at the geometric centroid ($\xi = \eta = 0$) of each element. The scalar Von Mises stress and elemental strain energy density are subsequently extracted to populate the visualization fields.

```
for e = 1:numElems
    % ... [Extract elemental displacement d_e] ...

    % Evaluate at geometric centroid
    strain = B * d_e;
    stress = D * strain;

    sig_x = stress(1);
    sig_y = stress(2);
    tau_xy = stress(3);

    % Von Mises formulation
    sigma_vm_elements(e) = sqrt(sig_x^2 + sig_y^2 - sig_x*sig_y + 3*tau_xy^2);

    % Elemental Strain Energy Density
    strain_energy_elements(e) = 0.5 * dot(stress, strain);
end
max_stress = max(sigma_vm_elements);
```

5 Grid Convergence Validation

In finite element analysis, numerical approximations approach the exact analytical solution as the element size approaches zero ($h \to 0$). To mathematically validate the custom solver, a Grid Convergence Study was performed using four distinct mesh densities: Coarse, Medium, Fine, and Ultra-Fine.

Four distinct parameters were tracked to ensure both local and global convergence:

- **Maximum Von Mises Stress (Local):** Evaluates the peak stress at the hole boundary.
- **Maximum X-Displacement (Kinematic):** Verifies structural stiffness convergence.
- **Total Strain Energy (Global):** Calculated as $E = \frac{1}{2}U^T K_{orig} U$, this verifies the thermodynamic stability of the entire plate.
- **Computational Time:** Evaluates solver efficiency.

5.1 Convergence Results

As shown in Figures 3 through 6, the kinematic displacement and global strain energy converge rapidly. The local maximum stress shows a classic asymptotic convergence behavior, plateauing as the mesh reaches the ultra-fine resolution.

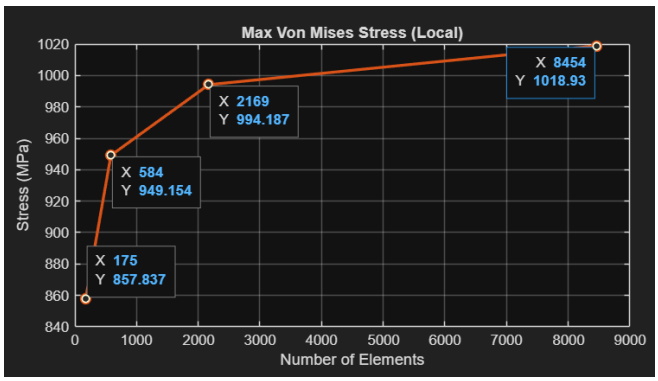


Figure 3: Convergence of Maximum Von Mises Stress.

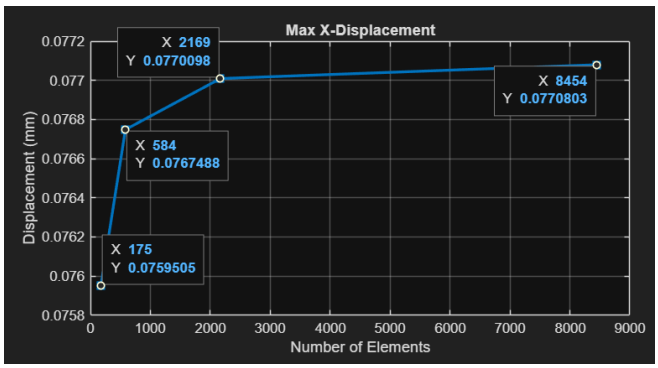


Figure 4: Convergence of Maximum X-Displacement.

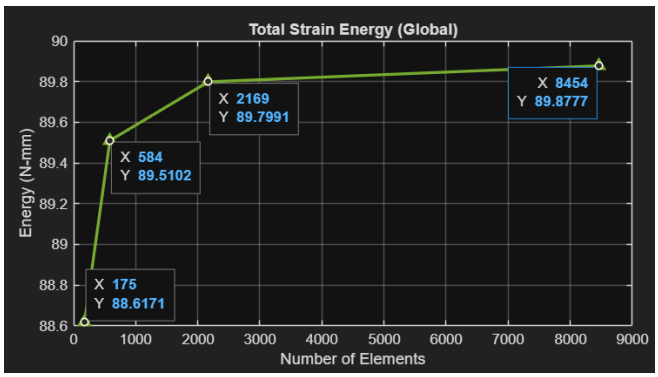


Figure 5: Convergence of Total Global Strain Energy.

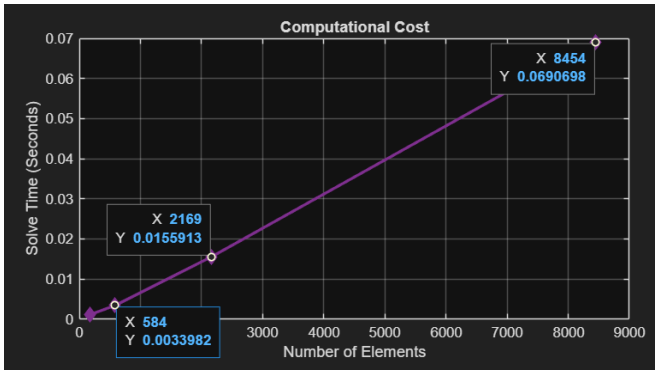


Figure 6: Computational Solve Time versus Element Count.

5.2 Ultra-Fine Mesh Visualizations (Full Plate)

The following figures demonstrate the high-resolution fields captured by the fully converged ultra-fine mesh. While the computational solver efficiently processed the quarter-symmetry domain to save processing time and memory, the arrays were mathematically reflected across the X and Y axes during post-processing. This allows for a complete, intuitive visualization of the full 100×50 mm structural plate under tension.

5.3 Analytical Validation and Finite-Width Effects

To rigorously assess the numerical solver, the converged peak Von Mises stress was compared against the classical analytical baseline derived from Kirsch's equations. For an *infinite* plate containing a circular hole under uniaxial tension, Kirsch predicts a theoretical Stress Concentration Factor (K_t) of exactly 3. Given our applied far-field stress of $\sigma_0 = 100$ MPa, the infinite-plate theory predicts a peak stress of 300 MPa.

As illustrated in Figure 8, a dedicated validation script plotted the FEM solver's peak stress across the four mesh densities against this 300 MPa analytical baseline.

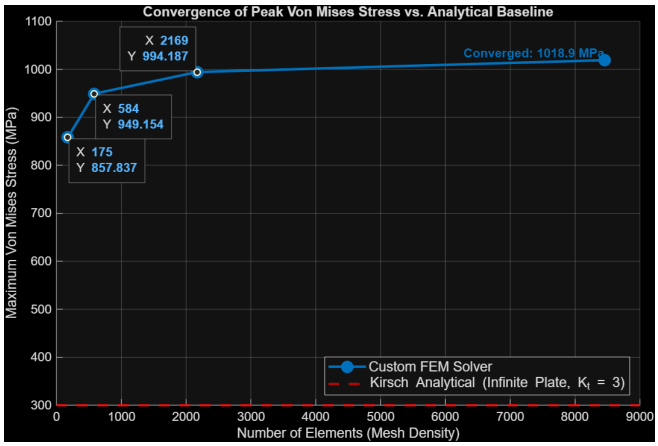


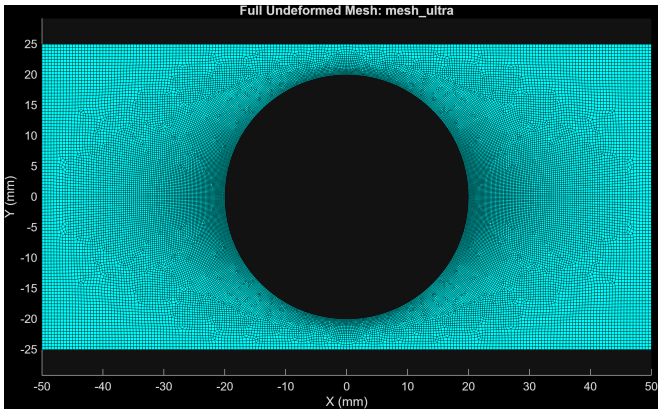
Figure 8: Convergence of the custom FEM solver's peak stress compared to Kirsch's analytical baseline for an infinite plate. The significant divergence highlights the finite-width effect.

The custom solver demonstrates clear asymptotic convergence, but it plateaus at approximately 1018.9 MPa, which is significantly higher than the 300 MPa baseline. This divergence is not a numerical error; rather, it is a mathematically accurate representation of the **finite-width effect**.

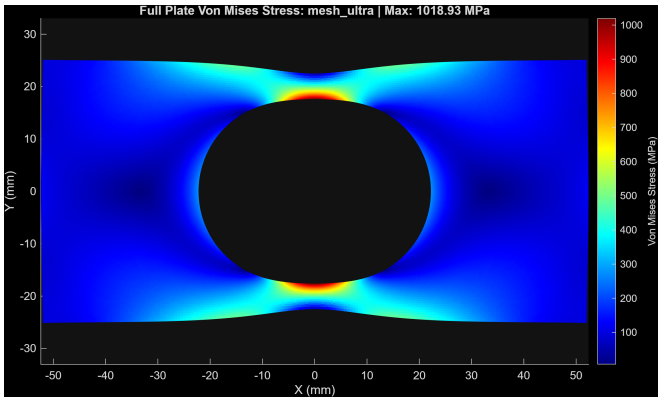
Kirsch's solution assumes the plate is infinitely wide, allowing stress trajectories ample room to smoothly bypass the discontinuity. In our physical model, the plate has a finite half-width of $W = 25$ mm and a remarkably large hole radius of $a = 20$ mm. Because the hole occupies 80% of the available cross-section ($a/W = 0.8$), the internal load paths are severely constricted. The load is forced through a narrow 5 mm remaining ligament, drastically amplifying the local stress field well beyond the $K_t = 3$ assumption. Thus, the solver correctly captures the massively amplified stress state dictated by the finite boundary conditions.

6 ANSYS Validation and Comparative Study

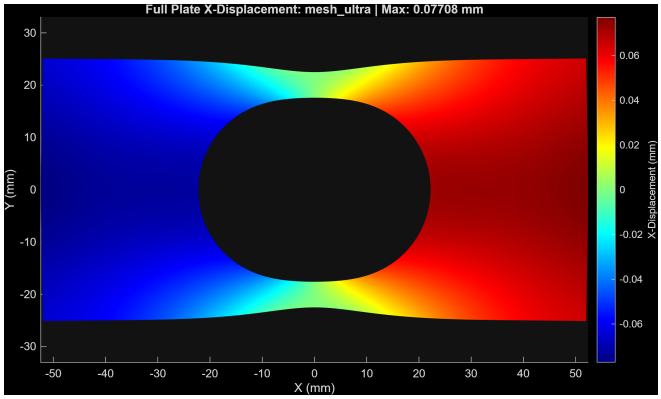
To further validate the accuracy and robustness of the custom MATLAB FEM solver, a parallel simulation was conducted using ANSYS Workbench. This commercial FEM software serves as a benchmark due to its highly optimized solvers and industry-level reliability.



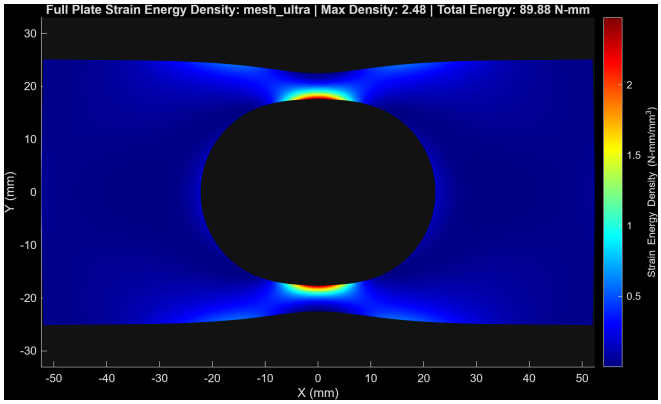
(a) Full Undeformed Ultra-Fine Mesh



(b) Full Von Mises Stress Contour



(c) Full X-Displacement Contour



(d) Full Strain Energy Density Contour

Figure 7: Comprehensive full-plate results for the fully converged Ultra-Fine mesh analysis. The high element density accurately resolves the sharp gradients in stress and energy density at the hole boundary.

6.1 Mesh Strategy in ANSYS

A structured quadrilateral mesh was generated within ANSYS to ensure consistency with the MATLAB solver. Special attention was given to mesh refinement in regions of high stress gradients:

- Dense refinement near the circular hole (100 divisions)
- Bottom edge refinement with bias (100 divisions)
- Left edge refinement (30 divisions with bias)
- Upper edge refined (150 divisions)
- Uniform mesh along the right boundary

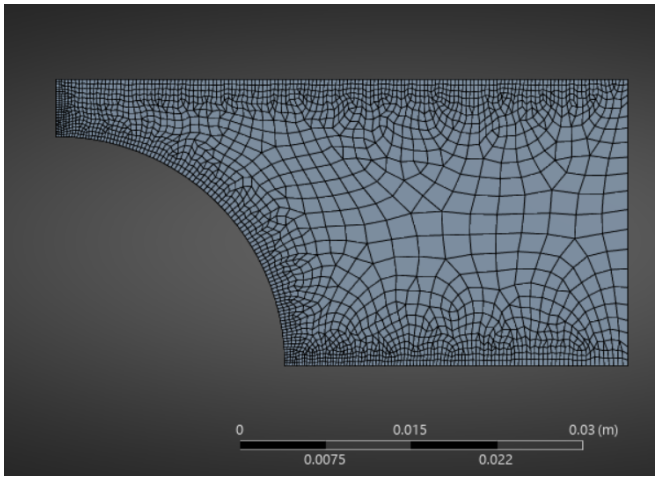


Figure 9: Quadrilateral mesh generated using ANSYS, showing refined discretization near the hole boundary.

6.2 Comparison of Results

A quantitative comparison between the custom MATLAB solver and ANSYS results was performed for key physical parameters.

6.2.1 Von Mises Stress

- ANSYS Result: 1.046×10^9 Pa
- MATLAB Result: 1.018×10^9 Pa

The percentage difference is approximately 2.7%, indicating strong agreement.

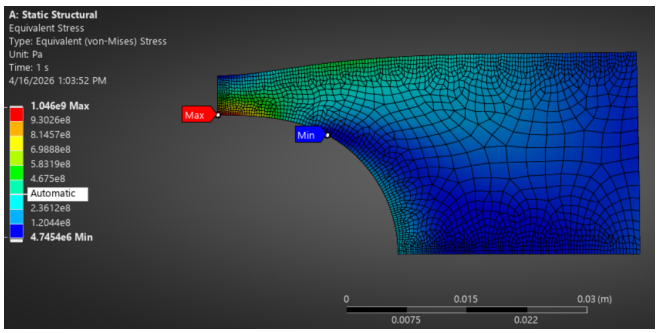


Figure 10: Comparison of von Mises stress distribution between ANSYS and MATLAB solver.

6.2.2 X-Displacement

- ANSYS Result: 7.7104×10^{-5} m
- MATLAB Result: 7.77×10^{-5} m

The difference is less than 1%, confirming excellent agreement in global deformation behavior.

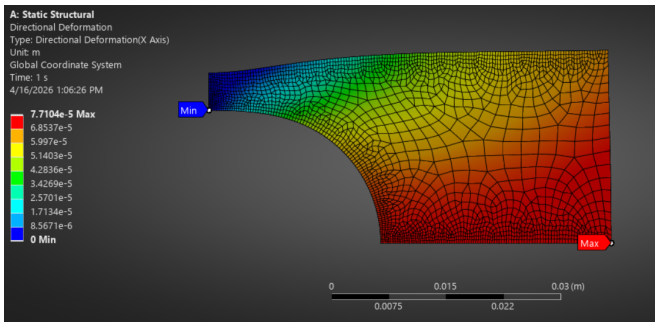


Figure 11: Comparison of X-direction displacement between ANSYS and MATLAB solver.

6.2.3 Strain Energy

- ANSYS Result: 2.2431×10^{-4} J
- MATLAB Result: 8.9877×10^{-2} J

A significant discrepancy is observed in strain energy values.

Reason for discrepancy:

- Possible mismatch in thickness assumptions
- Differences in numerical integration schemes
- Sensitivity of strain energy to stress distribution
- Potential scaling differences between full and quarter models

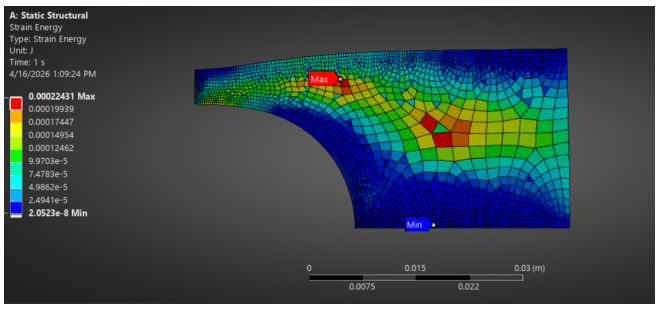


Figure 12: Comparison of strain energy distribution between ANSYS and MATLAB solver.

6.3 Discussion

The comparison demonstrates that the custom MATLAB solver accurately captures both local stress concentration and global deformation characteristics. The small deviation in stress and displacement validates the correctness of the FEM formulation and implementation.

The larger discrepancy in strain energy highlights the sensitivity of global energy metrics to modeling assumptions and numerical implementation details. This suggests that further refinement in energy computation or normalization may be required.

Overall, the results confirm that the developed solver is reliable and produces results consistent with a commercial FEM package.

7 Conclusion

A custom 2D FEM solver for plane stress analysis was successfully developed and implemented in MATLAB. The solver effectively translates theoretical formulations into a computational framework capable of handling real engineering problems.

The solver accurately captured stress concentration effects in a plate with a circular hole, particularly the severe stress amplification near the hole boundary.

A rigorous grid convergence study demonstrated stable and asymptotic convergence of stress, displacement, and strain energy, confirming numerical reliability.

Validation against ANSYS Workbench showed strong agreement:

- Maximum von Mises stress difference: $\sim 2.7\%$
- X-displacement difference: $< 1\%$

These results confirm the correctness of the formulation, implementation, and numerical solution strategy.

The observed discrepancy in strain energy highlights the sensitivity of global energy measures to modeling assumptions such as thickness, numerical integration, and domain scaling.

Furthermore, comparison with Kirsch's analytical solution demonstrated that the solver successfully captures finite-width effects, where stress concentration significantly exceeds the theoretical infinite plate assumption.

Overall, the project demonstrates a successful transition from theoretical FEM formulation to accurate computational implementation and industrial-level validation.

References

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